

VLM-Guided Failure Localization and Verified Counterfactual Hindsight

for Sparse-Reward Manipulation

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CS 285 · UC Berkeley · Spring 2026

A failure-direction, per-timestep, IS-grounded alternative to VLM-RB.

Problem statement

- Sparse-reward goal-conditioned RL on Fetch (**Push, PickAndPlace, Slide**) — reward fires only on success.
- Algorithm: **SAC + replay buffer**. Standard credit-assignment tools (TD-error PER, HER) treat all transitions as opaque.
- Core question: **which transitions deserve more replay weight?** We propose: the ones the VLM identifies as the *cause of failure*.

Concurrent work — Sharony et al. (Feb 2026, NVIDIA + Technion)

- **VLM-RB (arXiv:2602.01915)**: frozen VLM scores 32-frame sub-trajectory clips for *goal satisfaction* — success direction.
- **Mixture-with-uniform** at sampling: $\lambda_t \cdot \mu_{\text{VLM}} + (1 - \lambda_t) \cdot \mu_U$. Requires λ warm-up.
- Evaluated on **MiniGrid + OGBench**. No Fetch / Gymnasium-Robotics. No oracle baseline.
- Reports 11–52% success gain, 19–45% sample-efficiency gain over PER/UER.

THREE MONTHS AHEAD OF US. WE MUST DIFFERENTIATE.

Our 2-track approach

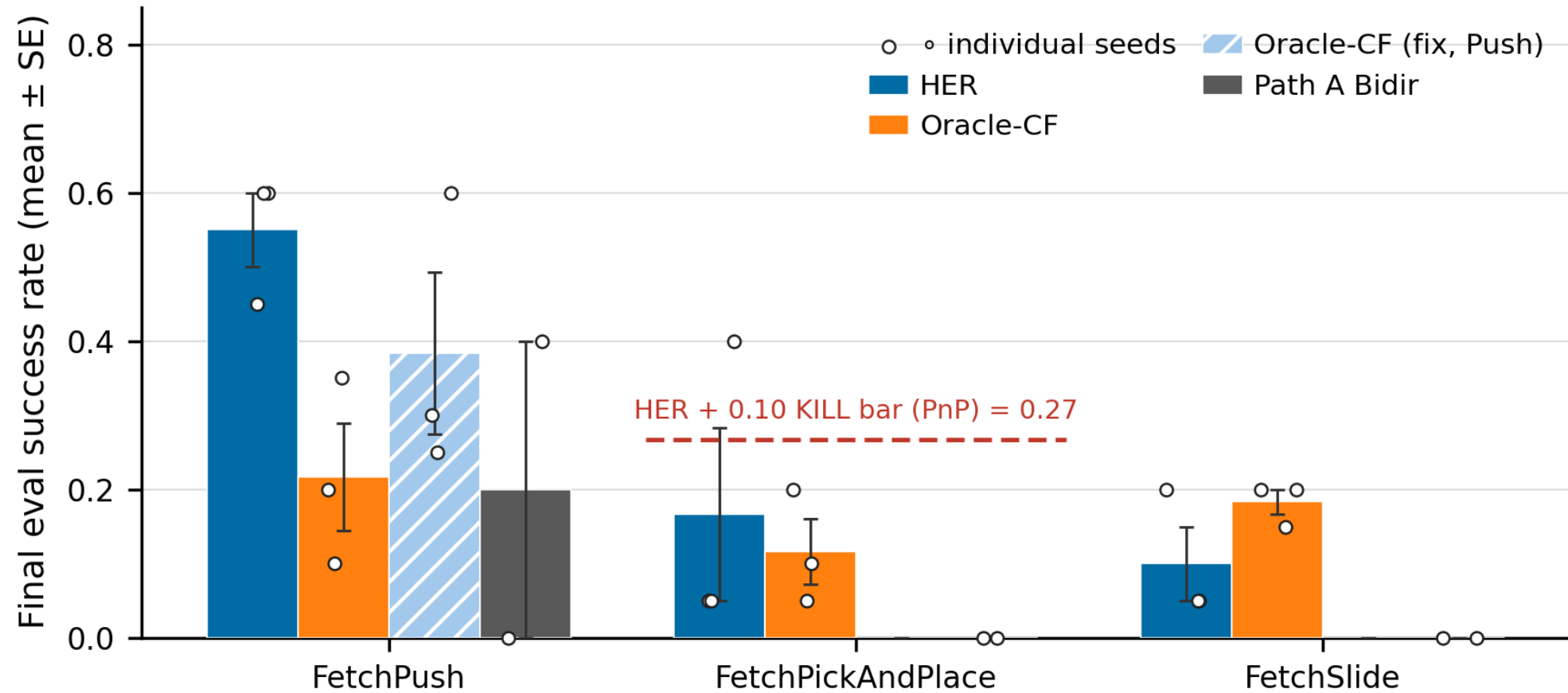
Path A — Failure-Direction PER

- **Semantic PER:** VLM localizes failure timestep; PER priority gets multiplicative window boost.
- **Oracle v3** (contact-aware): ballistic → contact-loss → argmin; decouples mechanism from VLM accuracy.
- **Bidirectional buffer:** $\text{success_weight} \times \text{failure_weight}$; degenerates to PER when both are 1.
- **HER baseline sweep** (27 jobs on Modal, overnight).

Path C — Verified Counterfactual Hindsight

- **Counterfactual VLM:** at failure timestep, ask *what action should the agent have taken?*
- **Verified-CF (N1):** roll VLM's action in a sim fork; accept only if env's own sparse reward fires.
- **Relabel** with confidence 1.0 (env signed off). Teleport collapse structurally inexpressible.
- **Smoke:** 4/4 PASS, 17.6 ms per verification.

Headline result — overnight Path C kill experiment



Final eval success rate at 250k steps. HER vs Oracle-CF (pre-fix and post-fix Push) vs Path A Bidir. $n=3$ seeds for HER/OCF, $n=2$ for Bidir. Pre-registered HER+0.10 KILL bar drawn on PnP.

■ **KILL.** $\Delta(\text{OCF}-\text{HER})$ on FetchPickAndPlace = **-0.05** — well below the +0.10 pre-registered threshold. Post-fix Push $\Delta = -0.17$. Path A Bidir $5/6 = 0.0$. HER on Fetch is at the credit-assignment ceiling; bottleneck is exploration, not credit.

Theoretical motivation (N3): Semantic PER as IS posterior reweighting

$$\mu_{\text{Sem}}(\mathbf{i}) \propto \mu_{\text{P}}(\mathbf{i}) \cdot w_{\text{sem}}(\mathbf{i}; \tau)$$

where $w_{\text{sem}}(\mathbf{i}; \tau) = E_{t^* \sim q_{\phi}} [K_{\mathcal{W}}(\mathbf{i}; t^*)]$ — VLM is the proposal $q_{\phi}(t^* | \tau)$; priority factorises into learning-progress $\times$ causal-influence.

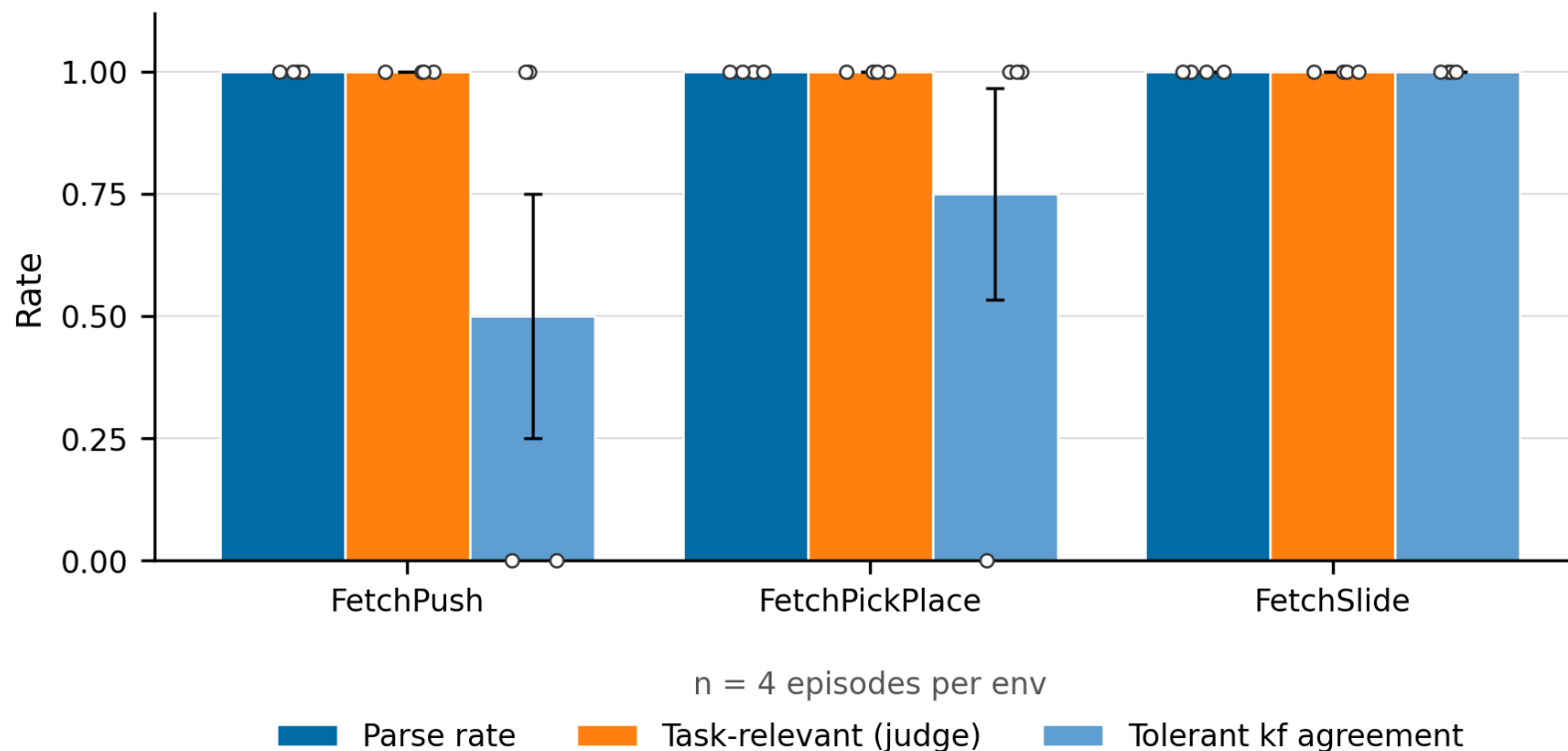
- **No λ warm-up:** multiplicative weight degenerates to PER cleanly when $w_{\text{sem}} = 1$.
- **Single hyperparam** (w_{max}) vs Sharony's mixture schedule.
- **Principled ablations:** ESS of w_{sem} ; $\text{KL}(p_{\text{oracle}} \parallel q_{\phi})$; IS-correction variants — diagnostics Sharony's recipe cannot produce.
- **Reframes the paper:** from "*VLM-PER beats PER by X%*" to "*foundation-model priors as credit-assignment oracles*" — a thesis, not a benchmark.

Verifiable counterfactuals (N1) — closes the teleport-collapse loophole

1. VLM	2. Sim fork	3. Reward check	4. Buffer
Propose corrective action at s_k	Restore qpos / qvel / mocap to s_k ; roll 50 steps	Env's own sparse reward fires? → accept	Relabel with confidence 1.0; HER fallback on reject

- **Teleport is structurally inexpressible** as a 4-vector action — failure mode eliminated by construction.
- **Off-policy correctness:** value function trains on the *real* sparse reward, not a VLM surrogate. Reward shaping doesn't get this guarantee.
- **Smoke test (N1_smoke_results.json):** 4/4 PASS, 17.6 ms per verification, ~0.4% wall-clock overhead.
- **Buffer integration:** ~50 LOC subclassing C2's CounterfactualHERBuffer stub.

Cross-task transferability (N2)



VLM keyframe-agreement with heuristic oracle. Same prompt, three envs. Claude Opus 4.7, K=5 keyframes, 4 episodes/env.

- **12/12 task-relevant** annotations across Push, PickAndPlace, Slide — *one prompt template, three envs, no retraining.*
- **Sharony cannot make this claim** — their architecture trains per benchmark (DQN/IQN on MiniGrid; SAC/TD3 on OGBench).

Honest publishability claim (post-overnight)

- **We have:** IS-posterior *framing* (§3), verifier *mechanism* (§4 / N1), cross-task *signal* (N2: 12/12 task-relevant), and a **pre-registered, falsified** Path C kill experiment — a real negative result, not noise.
- **We don't have:** a positive empirical headline on Fetch. Privileged Oracle-CF did not beat HER by 0.10 on any env; Path A Bidir 5/6 = 0.0; Phase 2 Modal sweep never fired (A1's HER sweep holding GPUs).
- **Recommended path:** reframe as 'methodology + theory + honest negative empirical' NeurIPS submission; scale N2 to n=50/env locally; let Phase 2 land as §5 ablation when Modal frees. Do not pivot to harder envs now — 4 days of training is a bet we cannot underwrite.
- **Realistic submission:** NeurIPS main track with negative-result framing OR NeurIPS workshop on Robot Learning as defensive fallback.

What got done overnight (22:00 → 10:00 PDT)

- **Path C kill experiment** ran to completion: 22 runs at 250k steps. $\Delta(\text{OCF-HER}) \text{ PnP} = -0.05 \rightarrow$ **KILL verdict**.
- **oracle_cf_push midpoint bug** identified and fixed (*ccb63d4*); 3-seed rerun: mean 0.383 vs HER 0.550 — verdict held.
- **Watchdog pivot fired** at 03:02 PDT; Path A bidir relaunch debugged (*eb899be*: missing config + wrong CLI form); 6 runs ran 03:33–06:01.
- **Path A Bidir result**: 5/6 = 0.0 success; only Push s42 = 0.4 at 300k. Bidir does not save Path A.
- **Paper edits**: §3 R2 honesty pass (*d2ce20b*); §6 Broader Impact restructured (*c488c6b*); uniqueness claim dropped; bias bound added (*95d9cee*).
- **Phase 2 Modal blocked all night**: A1's HER sweep holding all 10 GPU slots; watchdog correctly never auto-launched.
- **Crons stalled ~03:53 PDT** when LLM session lapsed; bash watchdog stayed alive but couldn't compensate. Morning consolidation crons (06:33-07:36) missed.

Next steps — three options for Daniel to choose

- **(a) Pivot to harder envs (Adroit / MetaWorld / AntMaze).** 18 runs, ~4 days wallclock. $P(\text{at least one OCF beats HER}) \approx 50\%$. Risk: 4 of 5 remaining days consumed.
- **(b) Reframe as methodology + theory + honest negative result paper.** No new training. 4 days for rewrite. Mitigation: workshop fallback.
- **(c) Scale N2 ($n=12 \rightarrow n=50/\text{env}$) + run Verified-CF Phase 2 when Modal frees.** 1 day local + 18h Phase 2. Phase 2 inherits Path C's empirical weakness.
- **Recommendation: (b) + a defensive slice of (c).** Reframe paper around IS-posterior + verifier + honest kill result. Scale N2 locally. Let Phase 2 land when Modal frees as §5 ablation. Do **not** pursue (a) right now — calendar is too tight.
- **Pre-Phase-2 must-fix:** *modal_app.py* WANDB_ENTITY regression (`_CODE_BLOCKER.md B2`) — every Phase 2 job will crash at `wandb.init` otherwise.